

Methodology: Electric Vehicle Charging Demand and Distribution Grid Congestion in California

Replication documentation for the **Distribution-Grid-EV-CA** pipeline
Based on R scripts in **R_Yanning/** (Li et al., UC Davis)

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Abstract

This document describes the end-to-end computational pipeline used to estimate spatially and temporally resolved electric vehicle (EV) charging demand on California distribution feeders, and to assess feeder overload relative to available headroom. The workflow integrates (i) California Statewide Travel Demand Model (CSTDm) trip tables, (ii) census geography and EV adoption scenarios from the EV Toolbox, (iii) empirical charging-session records, and (iv) utility feeder baseload and capacity data. Each processing stage is specified with input/output files, key variables, governing equations, and random seeds used for reproducibility.

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1 Introduction and study objective

The pipeline supports the analysis in *Impact of Electric Vehicle Charging Demand on Power Distribution Grid Congestion* (contact: Yanning Li, yspli@ucdavis.edu). The objective is to:

1. Assign EV adoption to synthetic households in CSTDM travel models across forecast years;
2. Derive EV trips and charging events by trip purpose (home, work, public);
3. Downscale charging events from traffic analysis zones (TAZs) to census blocks and distribution feeders;
4. Match events to empirical charging power profiles; and
5. Compare cumulative EV load against feeder headroom to quantify overload.

Scripts are numbered by stage: 01_xx–09_xx, where 08_xx and 09_xx produce figures and upgrade-cost estimates (not detailed here).

2 Conceptual framework

2.1 Geographic hierarchy

The model links four spatial units:

Unit	Role	Typical ID field
Census tract	EV Toolbox adoption; population weights	GE0IDtract / tract
Block group (BG)	Population share tract→BG; BG→TAZ map	GE0IDbg / GE0ID
TAZ	CSTDM HomeZone, trip origin I , destination J	TAZ / TAZ_Zone
Census block	Work/public event split; feeder assignment	GE0ID10
Distribution feeder	Load, capacity, overload	FeederID

2.2 Travel model components (CSTDM)

- **SDPTM** — Short-Distance Passenger Travel Model (intrazonal/interzonal daily trips).
- **LDPTM** — Long-Distance Passenger Travel Model (intercity trips; includes LDPTM_AccEgr access/egress legs).
- **ETM** — External Travel Model (trips entering California).

Light-duty vehicle (LDV) modes are SOV, HOV2, and HOV3.

2.3 Charging location types

Trips are tagged with a charging-purpose category **ChargeType**:

- **H** — Home;
- **W** — Work;
- **P** — Public (other activities).

2.4 Forecast horizon

Unless noted otherwise, scripts loop over years $y \in \{2022, \dots, 2045\}$ (24 years). EV household sampling uses random seed 42; charging-location draws use seed 12; block splitting uses seeds 66, 99; empirical session matching uses seed 36.

2.5 Pipeline flow

Stage 1: Extract LDV household home TAZs from CSTDM (01_01).

Stage 2: Compute block-group population shares within tracts (02_01).

Stage 3: Distribute tract-level EV/all households to BGs and TAZs (02_02).

Stage 4: Sample new EV households per TAZ per year (02_03).

Stage 5: Filter EV trips and assign charge types (02_04).

Stage 6: Generate charging events: SD survey-based; LD/ETM route-based (03_01, 03_02).

Stage 7: Block-TAZ population/job shares (04_01); split events to blocks (04_02); map blocks to feeders (04_03).

Stage 8: Match empirical sessions (05_02); aggregate hourly profiles (06_01).

Stage 9: Combine with grid baseload/capacity (07_05).

3 Stage 01: LDV household home zones

3.1 Script

01_01_get_LDV_households_per_TAZ.R

3.2 Purpose

Build pools of unique synthetic households (**SerialNo**) and their home TAZ (**HomeZone**) for each CSTDM component.

3.3 Inputs

Source	Description
CSTDM/SDPTM/*.csv	All trip files in SDPTM directory
LDPTM/LDPTM_Trips.csv	Long-distance trips
LDPTM/LDPTM_AccEgr.csv	Access/egress legs
ETM/trips_Ext.csv	External trips

3.4 Filters

Only light-duty auto trips are retained; each row below states which records are kept for that CSTDM component.

$$\text{SDPTM, LDPTM trips: } \text{Mode} \in \{\text{SOV}, \text{HOV2}, \text{HOV3}\} \quad (1)$$

Short- and long-distance main legs include single-occupant and high-occupancy vehicle modes only.

$$\text{LDPTM AccEgr: } (\text{DPurp} = \text{Access} \wedge \text{AccMode} = \text{Drive_Park}) \quad (2)$$

$$\vee (\text{OPurp} = \text{Egress} \wedge \text{EgrMode} = \text{Rent_Car}) \quad (3)$$

Access/egress legs are kept when they represent drive-park access or rental-car egress to the long-distance trip.

$$\text{ETM: } \text{ActorType} \in \{\text{CarLong}, \text{CarLocal}\} \quad (4)$$

$$\wedge \text{Mode} \in \{\text{SOV}, \text{HOV2}, \text{HOV3}\} \wedge \text{DPurp} = \text{I} \quad (5)$$

External trips are limited to inbound car trips by light-duty modes.

$$(6)$$

3.5 Outputs

```
data/mobility_data/CSTDM_processed/SDPTM_hh_home_TAZ.csv
data/mobility_data/CSTDM_processed/LDPTM_hh_home_TAZ.csv
data/mobility_data/CSTDM_processed/LDPTM_AccEgr_hh_home_TAZ.csv
data/mobility_data/CSTDM_processed/ETM_hh_home_TAZ.csv
```

Columns: SerialNo, HomeZone.

4 Stage 02: EV adoption, geography, and trips

4.1 Stage 02_01: Tract to block-group population share

Script: 02_01_map_tract_to_bg_2010.R

4.1.1 Inputs

- Census Tract Population_2010_short.csv — columns GEOID, Population
- Census Block Group Population_2010_short.csv — same structure

4.1.2 GEOID parsing

Census full GEOIDs use prefix 1400000US; scripts extract numeric IDs:

$$\text{GEOIDshort} = \text{numeric}(\text{substr}(\text{GEOID}, 11, \text{end})) \quad (7)$$

The block-group identifier drops the fixed census prefix and keeps the remaining digits.

$$\text{GEOIDtract (from BG)} = \text{numeric}(\text{substr}(\text{GEOID}, 11, \text{end} - 1)) \quad (8)$$

The parent tract ID is obtained by removing the last digit from the block-group GEOID.

$$(9)$$

4.1.3 Population share

For each block group b in tract t , the share is the block group's fraction of tract population (zero if the tract has no population):

$$\text{share}_{b,t} = \begin{cases} \frac{P_b}{P_t} & P_t > 0 \\ 0 & P_t = 0 \end{cases} \quad (10)$$

This share is later used to split tract-level household counts across block groups. Rows with invalid `tract_pop` strings are repaired by truncating trailing characters before recomputing shares.

4.1.4 Output

population share bg to tract 2010.csv: GEOIDbg, GEOIDtract, bg_pop, tract_pop, share.

4.2 Stage 02_02: EV household shares at TAZ

Script: 02_02_get_EV household share_per_bg_TAZ.R

4.2.1 Inputs

- households_all_by_tract.csv — tract, households
- evhouseholds_by_tract_acc2_2011.csv — tract, year, households
- population share bg to tract 2010.csv
- map bg to TAZ 2010.csv — GEOID (BG), TAZ_Zone

4.2.2 Tract to block group (EV households)

Replicate static shares for each year y :

$$EVhh_{b,y} = EVhh_{t,y} \times share_{b,t} \quad (11)$$

Tract-level EV households are allocated to each block group in proportion to its 2010 population share.

4.2.3 All households (tract total, then disaggregate)

$$hh_t = \sum_{\text{rows}} \text{households} \quad (12)$$

Total households in a tract are summed across input rows for that tract.

$$hh_b = hh_t \times share_{b,t} \quad (13)$$

All households are then disaggregated to block groups using the same population shares as for EV households.

$$(14)$$

4.2.4 TAZ aggregation

$$EVhh_{TAZ,y} = \sum_{b \in TAZ} EVhh_{b,y} \quad (15)$$

EV households are summed over all block groups mapped to the same TAZ.

$$hh_{TAZ} = \sum_{b \in TAZ} hh_b \quad (16)$$

All households in the TAZ are obtained by the same summation over member block groups.

$$EVhh_share_{TAZ,y} = \min\left(1, \frac{EVhh_{TAZ,y}}{hh_{TAZ}}\right) \quad (17)$$

The TAZ adoption rate is the EV-to-all household ratio, capped at 100%.

$$(18)$$

with $EVhh_share = 0$ if $hh_{TAZ} = 0$.

4.2.5 Outputs

- `evhh_tract_bg_ACC2.csv`
- `allhh_tract_bg.csv`
- `evhh_share_TAZ.csv` — `year`, `TAZ`, `EVhh_TAZ`, `hh_TAZ`, `EVhh_share`

4.3 Stage 02_03: Sample new EV households per TAZ

Script: `02_03_sample_EV hh.R`

4.3.1 California-wide share (ETM only)

$$\text{EVhh_share}_y^{\text{CA}} = \frac{\sum_{\text{TAZ}} \text{EVhh}_{\text{TAZ},y}}{\sum_{\text{TAZ}} \text{hh}_{\text{TAZ}}} \quad (19)$$

External-travel households use a statewide EV adoption rate computed as total EV households divided by total households across all TAZs.

4.3.2 Expected EV count per TAZ

For each home zone z and year y , with $n_{\text{hh},z}$ = count of LDV households in that TAZ:

$$n_{z,y}^{\text{EV}} = n_{\text{hh},z} \times \text{EVhh_share}_{z,y} \quad (\text{SD/LD/LD-AE}) \quad (20)$$

For in-state models, expected EV count equals the TAZ LDV pool times the local TAZ adoption share.

$$n_{z,y}^{\text{EV}} = n_{\text{hh},z} \times \text{EVhh_share}_y^{\text{CA}} \quad (\text{ETM}) \quad (21)$$

For external trips, the same pool is scaled by the California-wide share instead of a TAZ-specific share.

(22)

4.3.3 New EV households (year-over-year)

Within each TAZ z , ordered by year:

$$\Delta n_{z,y}^{\text{EV,new}} = n_{z,y}^{\text{EV}} - n_{z,y-1}^{\text{EV}}, \quad y > 2022 \quad (23)$$

After the first forecast year, new EVs are the increase in expected EV count relative to the prior year.

$$\Delta n_{z,2022}^{\text{EV,new}} = n_{z,2022}^{\text{EV}} \quad (24)$$

In 2022, all expected EV households in the TAZ are treated as newly adopting.

$$n_{z,y}^{\text{EV,new}} = \lfloor \Delta n_{z,y}^{\text{EV,new}} \rfloor \quad (25)$$

The number actually sampled is the non-negative integer part of the year-over-year increment.

(26)

4.3.4 Sampling without replacement

Function `sample.EVhh` (seed 42): for each year, sample $n_{z,y}^{\text{EV,new}}$ distinct `SerialNo` from remaining pool in TAZ z ; remove sampled IDs from pool for future years.

4.3.5 Outputs

```
EVhh_new_SDPTM_sample42.csv
EVhh_new_LDPTM_sample42.csv
EVhh_new_LDPTM_AccEgr_sample42.csv
EVhh_new_ETM_sample42.csv
```

Columns: HomeZone, hhID (=SerialNo), year.

4.4 Stage 02_04: EV trip extraction

Script: 02_04_get_EV_trips.R

4.4.1 Trip purpose → charge type mappings

SDPTM:

DPurp	ChargeType
O	H
W, P	W
S,H,T,C,L,R,K,Z	P

LDPTM: Bus/VFR→W; Rec→P; Com→W; Oth→P.

LDPTM AccEgr: adds Access→P.

ETM: all trips ChargeType = P; distance from TAZ matrix:

$$\text{Dist}_{\text{miles}} = \frac{\text{distance}_{\text{meters}}}{1609.344} \quad (27)$$

External-trip length is converted from network skims in meters to miles using the standard meters-per-mile factor.

4.4.2 Outputs (RDS)

```
EV_trips_new_SDPTM_sample42.rds, EV_trips_new_LDPTM_sample42.rds, EV_trips_new_LDPTM_AccEgr_sample42.rds,
EV_trips_new_EXT_sample42.rds.
```

Key trip columns: SerialNo, Person, Tour, Trip, DPurp, I, J, Mode, Dist, Time, year, ChargeType.

5 Stage 03: Charging events

5.1 Stage 03_01: Short-distance charging (SDPTM)

Script: 03_01_draw_charging_events_SD.R

5.1.1 Survey charging-location shares

Seven composite location patterns over {H, W, P} with probabilities:

H	W	P	Share
1	0	0	0.53
0	1	0	0.08
0	0	1	0.03
1	1	0	0.16
1	0	1	0.13
0	1	1	0.03
1	1	1	0.04

5.1.2 Tour-level procedure

1. Assign `TourID` per (`SerialNo`, `Person`, `Tour`).
2. For each tour, enumerate which of $\{H, W, P\}$ appear among trip purposes; form valid location patterns; normalize survey shares to tour-specific probabilities.
3. Draw one pattern per tour (seed 12).
4. A trip generates a charging event if its `ChargeType` matches an active location in the drawn pattern.
5. Sum trip distances by charging event; attach destination TAZ `J`, time, year.

5.1.3 Output

`charge_event_new_SDPTM_sample42_draw12.rds`: `TourID`, `chargeEvent`, `Trip`, `chargeDist`, `ChargeType`, `J`, `Time`, `year`.

5.2 Stage 03_02: Long-distance and external charging

Script: `03_02_charging_events_LD_ETM.R`

5.2.1 TAZ distance matrices

Parsed CSV folders: `parsed_100000_LDPTM`, `parsed_100000_ETM`. Columns: `from`, `to`, `distance`, `TAZ12`, (renamed `TAZway`).

Each OD pair (I, J) may have multiple rows (one per intermediate TAZ on path).

5.2.2 How this codebase generates data/mobility_data/TAZ_distance

The Python builder `build_taz_path_skims.py` creates both parsed folders directly from local GIS and CSTDM trip inputs.

Inputs

- Network links: `data/shps/CSTDM_sf.gpkg` with node IDs `A, B` and link weight `DISTANCE` (miles);
- TAZ polygons: `data/shps/taz_id.shp` (field `TAZ12`);
- TAZ centroids: `data/shps/TAZ_centroid_sf.gpkg` (fallback to polygon centroids if unavailable);
- OD pairs from CSTDM trips:
 - LDPTM: unique (I, J) from `LDPTM_Trips.csv`;
 - ETM: unique inbound (I, J) where `DPurp='I'` from `trips_Ext.csv`.

Algorithm

1. Build a directed graph from links using `DISTANCE` as edge weight (miles).
2. Derive node coordinates from line endpoints and snap each TAZ centroid to its nearest network node (KD-tree).
3. Assign each directed link to a TAZ by midpoint spatial join (`within`), then nearest-TAZ fallback for unmatched links.
4. For each origin zone, run one Dijkstra shortest-path tree and reconstruct node paths to all required destinations.
5. Convert path length to meters:

$$\text{distance}_{\text{meters}} = \text{distance}_{\text{miles}} \times 1609.344 \quad (28)$$

Output distance is stored once per OD pair and repeated for each path row.

6. Convert node path to TAZ sequence by mapped link TAZs, dropping consecutive duplicates, and forcing first/last TAZ as origin/destination.

ETM external-zone handling ETM origins *I* are external IDs (1–51), not California internal TAZ IDs (100+). The builder maps each ETM origin to a California entry TAZ using the ETM `Int` region label (with optional (*I*, *J*) overrides), then routes from that entry TAZ to destination *J* while keeping output `from=I` in the parsed file.

Outputs and run commands

```
python build_taz_path_skims.py --trip-set ld
python build_taz_path_skims.py --trip-set etm
python build_taz_path_skims.py --trip-set both
```

Outputs are chunked CSVs (`part_00001.csv, ...`) under:

```
data/mobility_data/TAZ_distance/parsed_100000_LDPTM/
data/mobility_data/TAZ_distance/parsed_100000_ETM/
```

Practical caveat Distances and pair coverage can differ from externally supplied route tables (for example `all_routes.csv` or `parsed_100000_ext.csv`) if those were produced from a different network snapshot, centroid connector definition, or external-zone connector logic.

5.2.3 Saved approach: Option 4 (LDPTM)

For each trip with distance D (miles):

$$n_{\text{charge}} = \lceil D/100 \rceil \quad (29)$$

The model assumes one charging stop roughly every 100 miles, rounded up to an integer count.

$$n_{\text{TAZ}, \text{trip}} = \text{number of path TAZs for } (I, J) \quad (30)$$

This is how many TAZ rows appear on the shortest-path skim between origin I and destination J .

$$q_{\text{cut}} = 1 - \frac{n_{\text{charge}}}{n_{\text{TAZ}, \text{trip}}} \quad (31)$$

The quantile cutoff selects the busiest path TAZs so that only n_{charge} locations remain after filtering.

$$(32)$$

Let f_k = frequency of TAZ k on network paths. Keep TAZs with $f_k > Q_{q_{\text{cut}}}(f \mid \text{trip})$ (per-trip quantile). Charging distance per retained TAZ:

$$\text{chargeDist} = \frac{D}{n_{\text{TAZ}, \text{charge}}} \quad (33)$$

Trip distance is split evenly across the TAZs that survive the frequency filter, giving miles attributed to each corridor charging event. `ChargeEventType` = `P` except at destination TAZ (`TAZway = J`), where `ChargeType` applies. Tag `other` $\in \{\text{corridor}, \text{destination}\}$.

5.2.4 ETM

Option 4 analogous (uses $\geq$ quantile for ETM). Saved output uses Option 4 for LD; ETM file may use Option 1 or 4 per script version—check saved RDS.

5.2.5 Outputs

charge_event_new_LDPTM_sample42.rds, charge_event_new_ETM_sample42.rds.

6 Stage 04: Spatial downscaling to blocks and feeders

6.1 Stage 04_01: Block shares within TAZ

Script: 04_01_get_share_block vs TAZ.R

$$\text{population_share}_c = \frac{\text{pop}_c}{\sum_{c' \in \text{TAZ}} \text{pop}_{c'}} \quad (34)$$

Each census block receives a fraction of TAZ population equal to its block population divided by the TAZ total.

$$\text{jobs_share}_c = \frac{\text{jobs}_c}{\sum_{c' \in \text{TAZ}} \text{jobs}_{c'}} \quad (35)$$

Workplace share uses LODS job counts the same way, to locate work and public charging events.

(36)

Blocks without LODS jobs: $\text{jobs}_c = 0$. Census block GEOID10 = last 14 characters of full block GEOID.

Output: share_block vs TAZ.csv.

6.2 Stage 04_02: TAZ events to blocks

Script: 04_02_split_charging_events_TAZ to block_yearly_new.R

Split home (H) events by `population_share`; work/public (W,P) by `jobs_share`.

For TAZ z , year y , total events $N_{z,y}$:

$$n_c^{\text{int}} = \lfloor N_{z,y} \cdot \text{share}_c \rfloor \quad (37)$$

Each block first receives the integer part of events proportional to population or jobs share.

$$R_{z,y} = N_{z,y} - \sum_c n_c^{\text{int}} \quad (38)$$

The residual counts how many events were lost to flooring and must be reassigned randomly.

(39)

Residual $R_{z,y}$ events assigned to $R_{z,y}$ randomly selected blocks with $\text{share}_c > 0$ (seed 66 for allocation; seed 99 for event ordering).

Outputs:

charge_event_new_block_home_sample42_draw12_split6699.rds
charge_event_new_block_workpublic_sample42_draw12_split6699.rds

6.3 Stage 04_03: Blocks to feeders

Script: 04_03_map_block to feeder.R

Combine PGE, SCE, SDGE feeder shapefiles. Assign blocks via (i) intersection with feeder lines (QGIS), (ii) nearest feeder for remaining blocks. Export combo shapefile/CSV for IOU blocks.

7 Stage 05: Empirical charging sessions

7.1 Script

05_02_sample_charging events_distance bins_IOU blocks.R

7.2 Energy demand bins

$$\text{demand (kWh)} = \frac{\text{dist}}{3} \quad (40)$$

Charging energy need is approximated as one third of trip or charge distance in miles, i.e. roughly 3 miles per kWh. Eight bins: (0, 5], (5, 10], ..., (80, ∞) kWh (same bins on empirical **energy**).

7.3 Charger type assignment

- Public: sample DC vs L2 by charger counts (8919 vs 29229); corridor LD trips forced to DC.
- Home: sample single-family vs multi-family (weights 56.4 vs 38.9).

7.4 Matching

For each (type, housing/level, bin) stratum, sample **eventID** from cleaned empirical pool **charging_session_all** (seed 36), matching count to events in stratum.

7.5 Output

session_feeder_new_bins_IOU_sample42_draw12_split6699_sample36.rds with **FeederID**, **year**, **start_hour**, **end_hour**, **charge_type**, **charger_level**, **power**, **energy**, **housing**.

8 Stage 06: Feeder charging profiles

8.1 Script

06_01_sum_charging profiles.R

For each event, expand to 24 hourly bins; assign load = **power** when hour ∈ [start_hour, end_hour] (wrap past midnight if needed). Sum by (**FeederID**, **charge_type**, **hour**, **year**).

Cumulative profile through year y :

$$L_{f,h,y}^{\text{cum}} = \sum_{y'=2022}^y L_{f,h,y'} \quad (41)$$

Cumulative feeder load at hour h in year y is the sum of that hour's EV charging power over all adoption years up to and including y .

Output: **profile_feeder_cum_bins_IOU_sample42_draw12_split6699_sample36.rds**.

9 Related: multi-day SOC-aware simulation (multiday_charging/)

A separate subproject (multiday_charging/) simulates fleet charging over 365 days without running CSTDM stages 03–04. Vehicles accumulate driving energy until a preferred charging interval elapses or the battery would deplete; sessions are drawn from the same empirical pool as stage 05. Full documentation: docs/multiday_charging_methodology.tex (companion PDF).

Key distinction: stages 03_01/03_02 decide *how many* charging events and *where* along trips; multiday_charging decides *when* drivers charge across days given pack size and driving, then samples empirical hourly shapes directly.

10 Stage 07: Grid overload assessment

10.1 Script

07_05_grid_EV.R

10.2 Grid quantities

Per feeder f , month m , hour h :

$$\text{capacity}_{f,m,h} = \text{baseload}_{f,m,h} + \text{headroom}_{f,m,h} \quad (42)$$

Feeder capacity is interpreted as existing baseload plus margin available for additional load.

$$\text{overload}_{f,m,h,y} = \text{EVload}_{f,h,y} - \text{headroom}_{f,m,h} \quad (43)$$

Overload measures how much EV demand exceeds headroom; positive values indicate congestion risk for that hour.

(44)

PGE/SCE/SDGE merged after utility-specific cleaning. EV load replicated across 12 months within each year.

Positive **overload** indicates EV demand exceeds available headroom for that hour.

Output: overload_sample42_draw12_split6699_sample36.rds.

11 Master data dictionary (selected fields)

Field	Meaning
SerialNo / hhID	Synthetic household identifier in CSTDM
HomeZone	Home TAZ
I, J	Trip origin/destination TAZ
DPurp	Detailed trip purpose code
ChargeType	Aggregated charging location: H, W, P
Dist	Trip distance (miles)
GE0IDbg	Numeric 2010 block-group ID
GE0ID10	14-digit census block ID
TAZ_Zone / TAZ	CSTDM traffic analysis zone
share	BG population fraction of tract
EVhh_share	EV households / all households in TAZ

Field	Meaning
chargeDist	Energy-relevant distance attributed to a charging event (mi)
FeederID	Utility distribution feeder/circuit ID
headroom	Capacity margin above baseload (kW)
overload	EV load minus headroom (kW)

12 Reproducibility: random seeds

Step	Seed
EV household sampling (02_03)	42
SD charging-location draw (03_01)	12
Block event count residual (04_02)	66
Block event ordering (04_02)	99
Empirical session matching (05_02)	36

13 Software and dependencies

R packages: `data.table`, `tidyverse`, `sf`. Python ports use `pandas`, `geopandas`, `numpy`. Large intermediate files stored as `.rds` (R) or `.pkl` (Python).

References

- California Statewide Travel Demand Model (CSTDm), Caltrans.
- U.S. Census Bureau, 2010 Summary File 1; TIGER/Line shapefiles.
- LEHD LODES (2019 workplace area characteristics).
- EV Toolbox, UC Davis / Strategic Growth Council.
- Li, Y., et al., *Impact of Electric Vehicle Charging Demand on Power Distribution Grid Congestion*.